GIDS Biomedical Data Science Hackathon 2024 – The Browns

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The Challenge

- Develop a model to identify associations between chromatin accessibility (ATAC-seq) and gene expression (RNA) using single cell multiome data

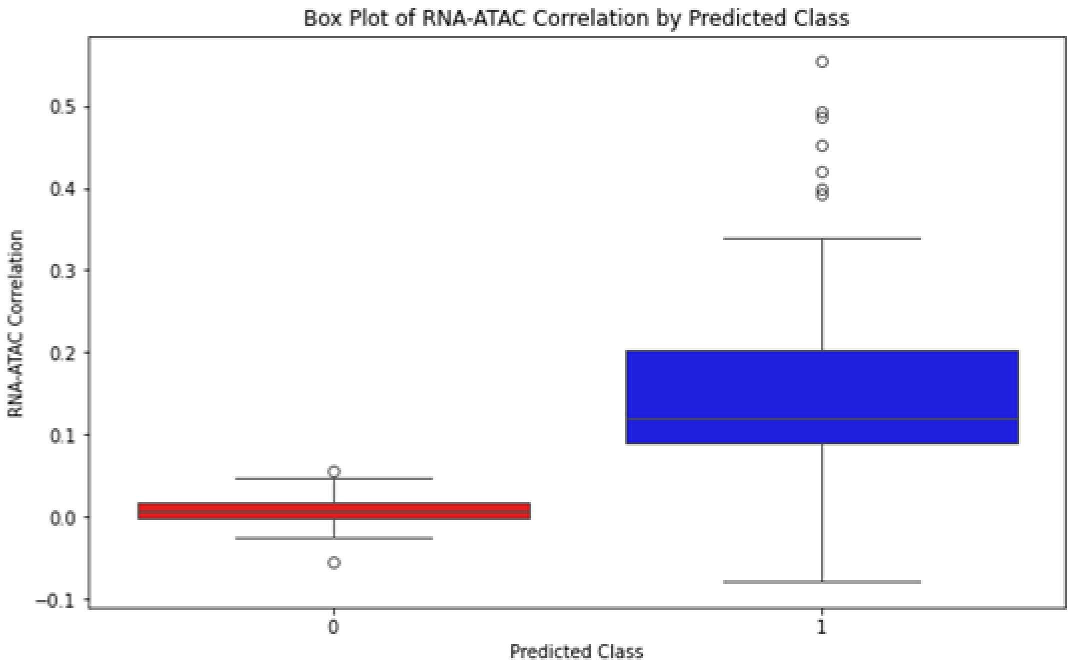
Our Approach

- Preprocess & clean data (convert to numeric)
- Compute average RNA & ATAC-seq counts (drop zero values), merge with training data based on gene
- Iterate through each pair in the training data, compute pairwise statistical correlation between mean RNA & ATAC-seq values, append to training data
- Repeat steps for testing dataset
- Implement XGBoost Machine Learning Classifier to predict Peak2Gene associations on testing data using mean counts & correlation

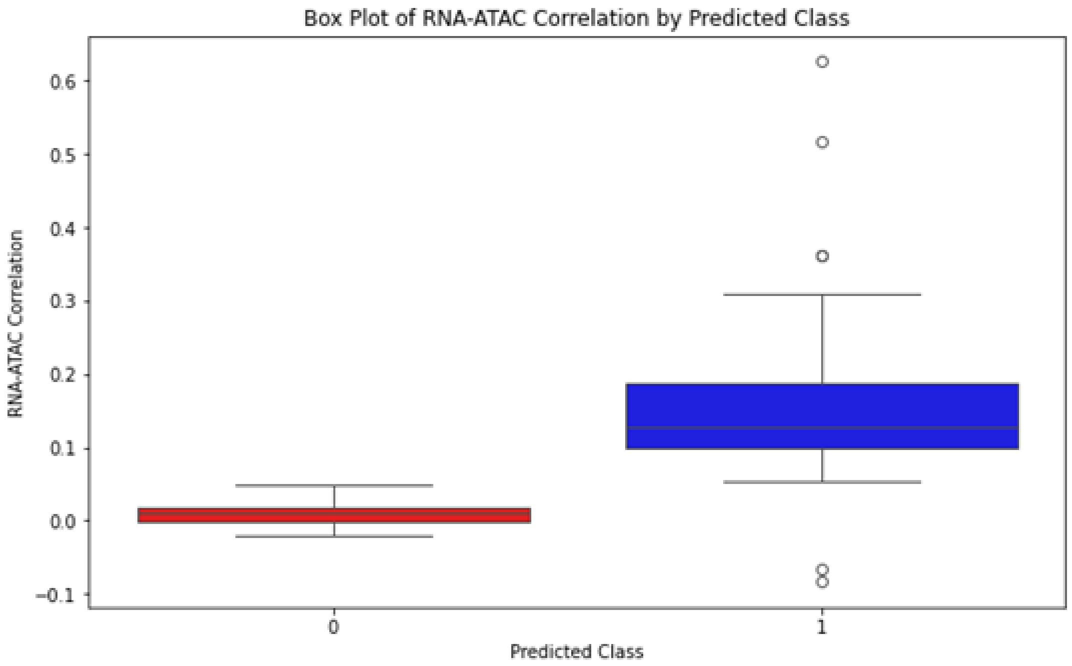
	peak	gene	Pair	Peak2Gene	mean_rna_counts	mean_atac_counts	rna_atac_correlation
0	chr1-89196985-89201657	GBP2	chr1-89196985-89201657_GBP2	1	2.103701	0.380127	0.083005
1	chr6-33077557-33083333	HLA-DPA1	chr6-33077557-33083333_HLA-DPA1	1	1.340447	0.691230	0.102732
2	chr6-137789753-137792920	TNFAIP3	chr6-137789753-137792920_TNFAIP3	1	0.675559	0.662888	0.118542
3	chr1-212604203-212626574	ATF3	chr1-212604203-212626574_ATF3	1	1.250417	6.757586	0.254281
4	chr2-96541661-96555628	ARID5A	chr2-96541661-96555628_ARID5A	1	0.273758	1.234078	0.085944

	peak	gene	Pair	Peak2Gene	mean_rna_counts	mean_atac_counts	rna_atac_correlation
0	chr1-1245493-1248050	SDF4	chr1-1245493-1248050_SDF4	?	0.150383	0.112371	0.002070
1	chr1-1330394-1334148	MRPL20	chr1-1330394-1334148_MRPL20	?	0.288096	0.180393	0.006696
2	chr1-2145904-2147150	FAAP20	chr1-2145904-2147150_FAAP20	?	0.127376	0.124708	-0.007857
3	chr1-9713011-9736481	PIK3CD	chr1-9713011-9736481_PIK3CD	?	1.222407	2.206069	0.146060
4	chr1-21287896-21301043	ECE1	chr1-21287896-21301043_ECE1	?	0.310103	2.377459	0.110263

RNA-ATAC Prediction of Training/Testing Data – Box Plot

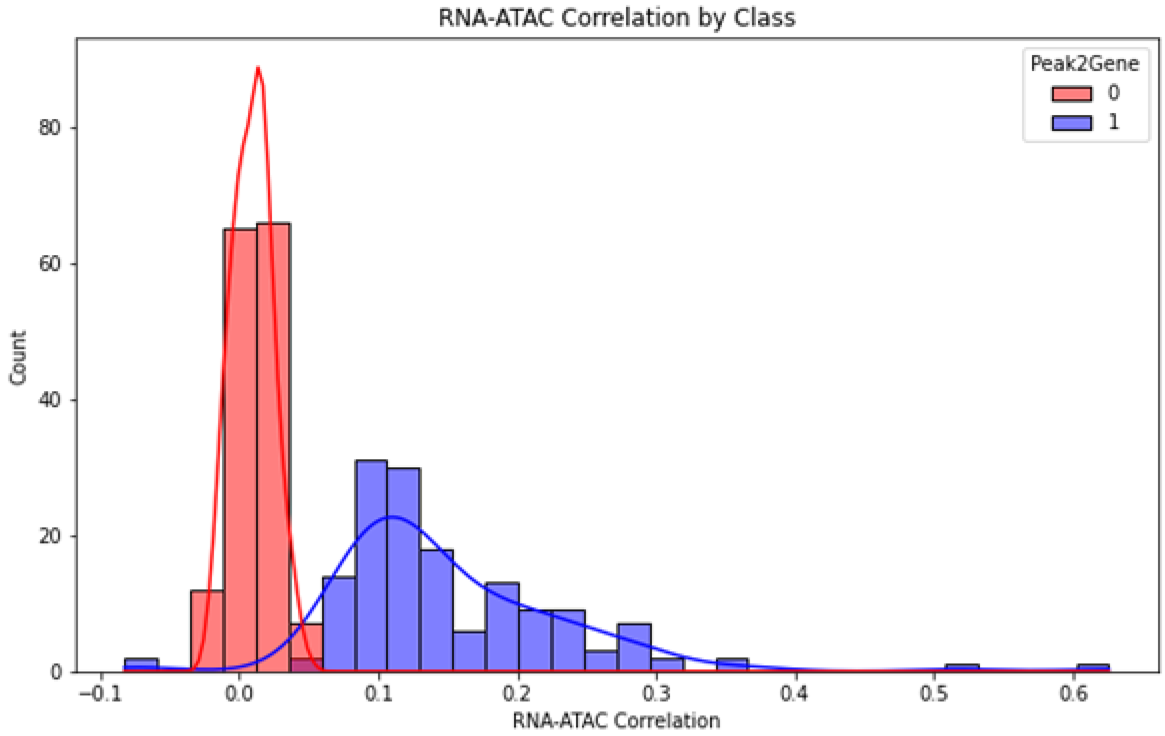


Training Set

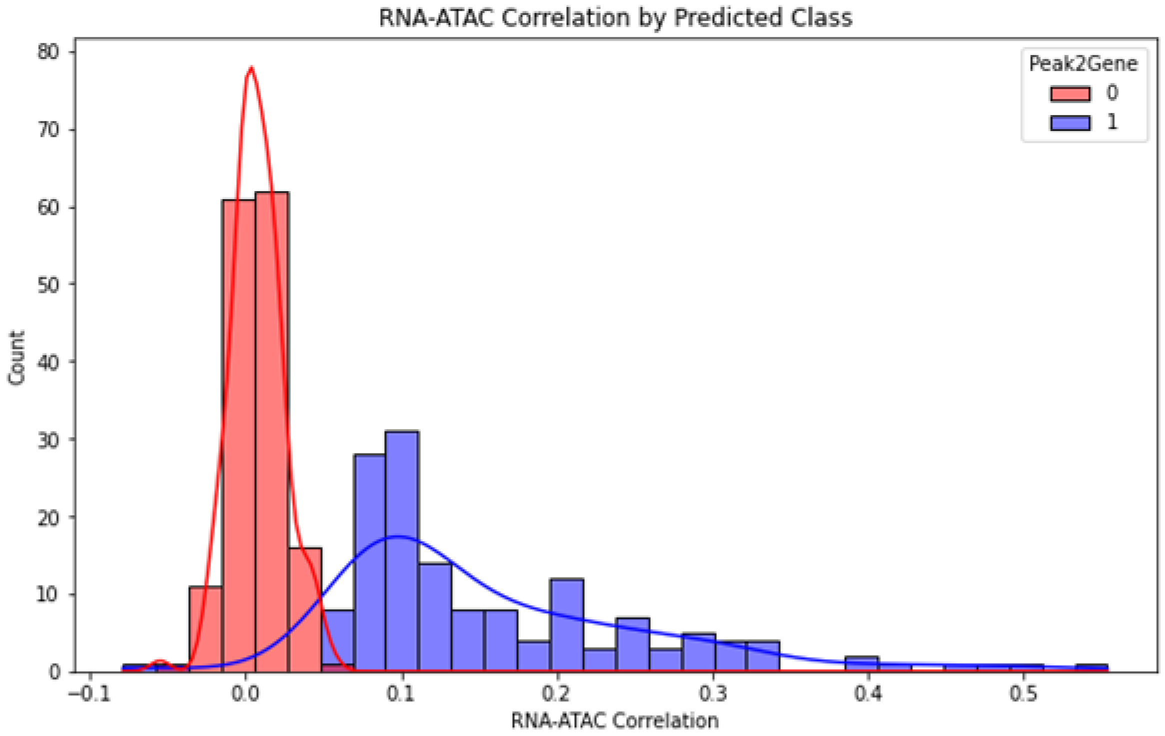


Testing Set

RNA-ATAC Prediction of Training/Testing Data – Histogram

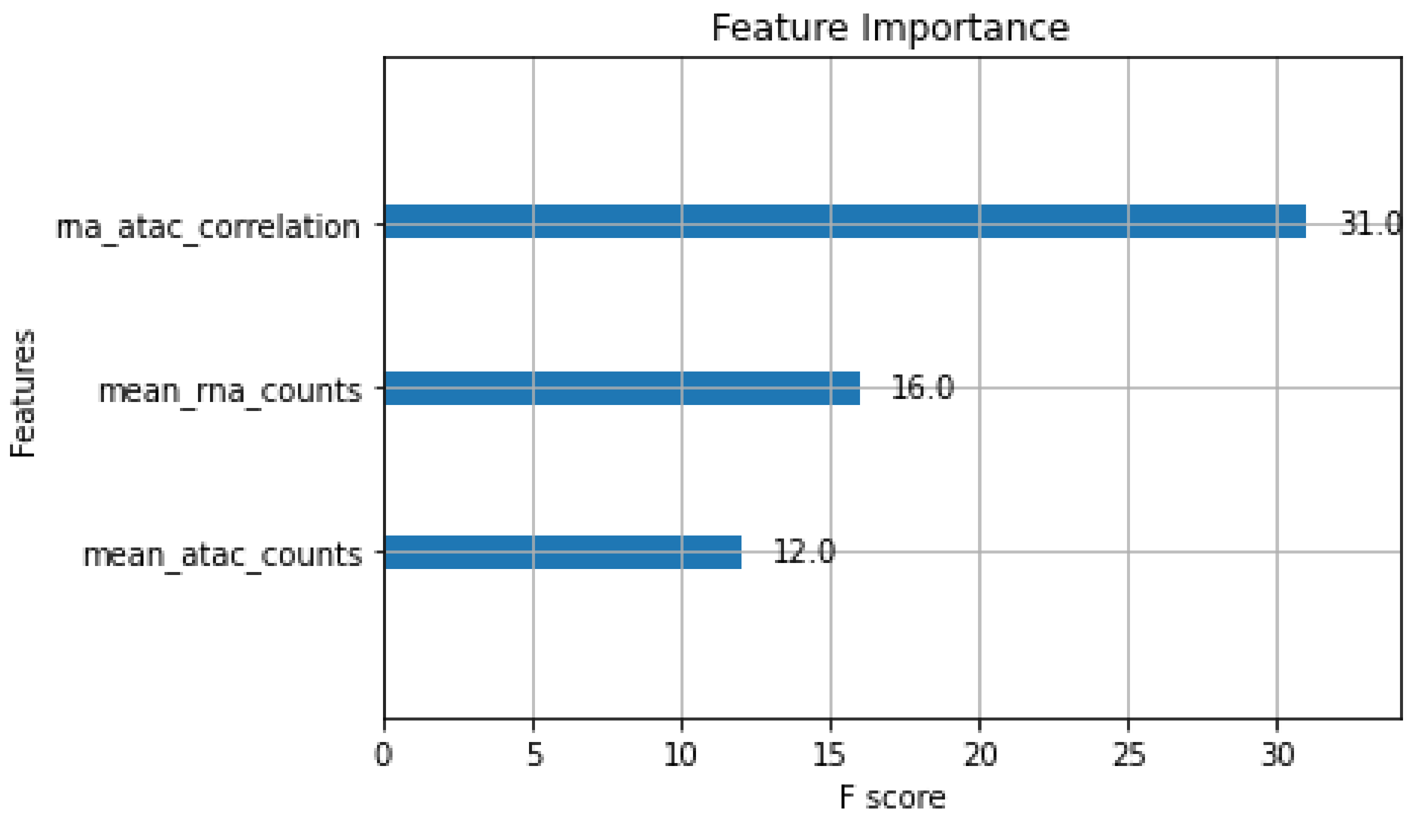


Training Set



Testing Set

Feature Importance



Results & Takeaways

- MCC score was calculated to 0.9868 (near 100% accuracy)
- Feature engineering: statistical correlation was found to be the most important feature in making predictions.
- XGBoost classifier is effective in making predictions, but other classifiers like Logistic Regression yielded the same results, implying feature importance supersedes prediction model.
- Additional Considerations: Adding correlation significance as a feature for analysis, handling outliers